

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab 09



Submitted to:

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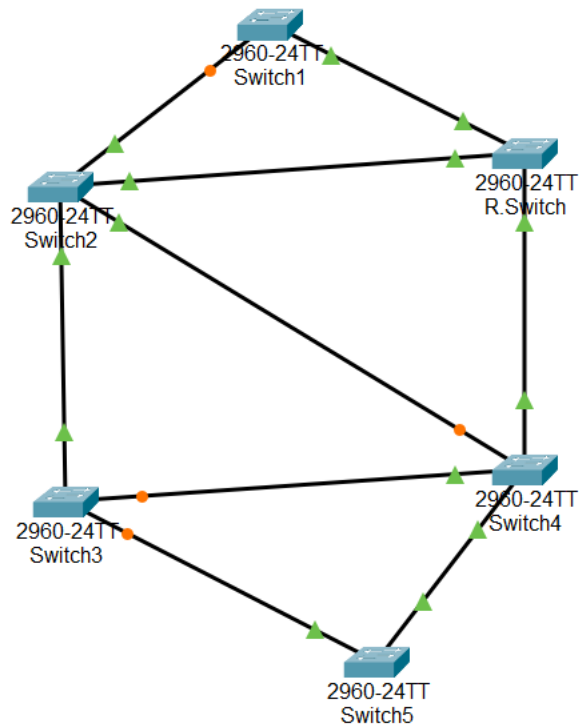
Subject: computer network

Department: BSSE-5-Afternoon

Date: 07-12-2024

Objectives

- Study the concept of STP in packet tracer and learn to configure the STP in switches



Enable Spanning Tree Protocol

```
Zulqarnain-SP21379-RW(config)#  
Zulqarnain-SP21379-RW(config)#  
Zulqarnain-SP21379-RW(config)#spa  
Zulqarnain-SP21379-RW(config)#spanning-tree v  
Zulqarnain-SP21379-RW(config)#spanning-tree vlan 1  
Zulqarnain-SP21379-RW(config)#  
Zulqarnain-SP21379-RW(config)#  
Zulqarnain-SP21379-RW(config)#
```

Set Bridge Priority: To force a specific switch as the root bridge:

```
Zulqarnain-SP21379-RW(config)#  
Zulqarnain-SP21379-RW(config)#  
Zulqarnain-SP21379-RW(config)#spa  
Zulqarnain-SP21379-RW(config)#spanning-tree v  
Zulqarnain-SP21379-RW(config)#spanning-tree vlan 1 p  
Zulqarnain-SP21379-RW(config)#spanning-tree vlan 1 priority 4096  
Zulqarnain-SP21379-RW(config)#  
Zulqarnain-SP21379-RW(config)#  
Zulqarnain-SP21379-RW(config)#
```

View Spanning Tree Status:

```

Zulqarnain-SP21379-RW#
Zulqarnain-SP21379-RW#sh
Zulqarnain-SP21379-RW#show spa
Zulqarnain-SP21379-RW#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID      Priority    4097
              Address     0001.64E5.B56C
              This bridge is the root
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID    Priority    4097 (priority 4096 sys-id-ext 1)
              Address     0001.64E5.B56C
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Desg FWD 19        128.1    P2p
Fa0/2          Desg FWD 19        128.2    P2p
Fa0/3          Desg FWD 19        128.3    P2p

Zulqarnain-SP21379-RW#
Zulqarnain-SP21379-RW#

```

Enable rapid spanning tree

```

Zulqarnain-SP21379-RW(config)#spa
Zulqarnain-SP21379-RW(config)#spanning-tree m
Zulqarnain-SP21379-RW(config)#spanning-tree mode ra
Zulqarnain-SP21379-RW(config)#spanning-tree mode rapid-pvst
Zulqarnain-SP21379-RW(config)#

```

Set inter-switch ports as trunk ports to allow VLAN traffic to propagate correctly:

```

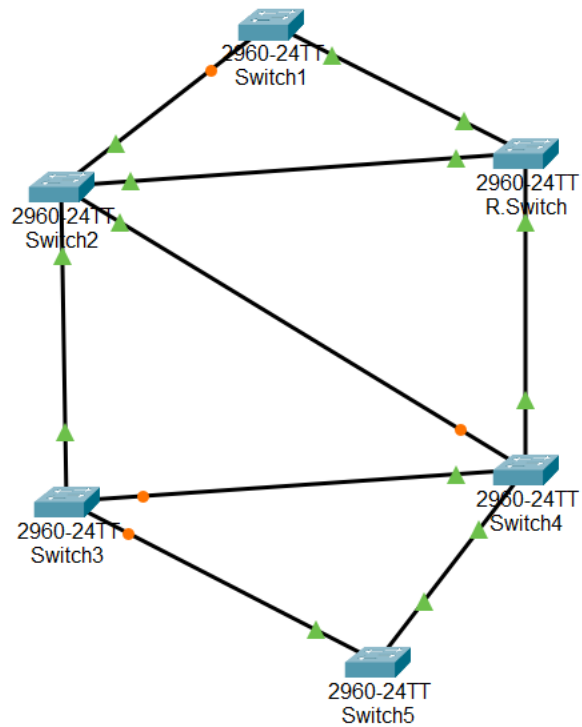
Zulqarnain-SP21379-RW(config)#inter
Zulqarnain-SP21379-RW(config)#interface R
Zulqarnain-SP21379-RW(config)#interface Range f0/1-3
Zulqarnain-SP21379-RW(config-if-range)#sw
Zulqarnain-SP21379-RW(config-if-range)#switchport m
Zulqarnain-SP21379-RW(config-if-range)#switchport mode t
Zulqarnain-SP21379-RW(config-if-range)#switchport mode trunk

Zulqarnain-SP21379-RW(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to
down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to
down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

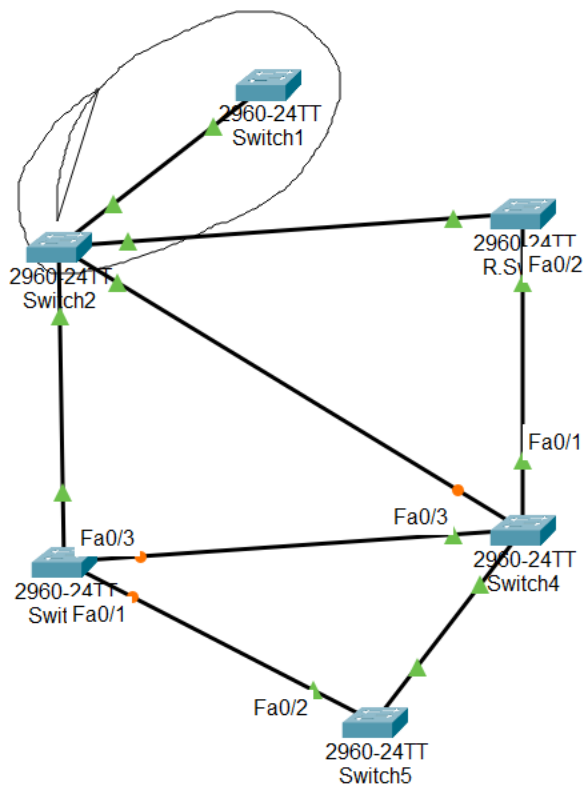
Zulqarnain-SP21379-RW(config-if-range)#
Zulqarnain-SP21379-RW(config-if-range)#

```

Disconnecting different switches



Disconnect Switch1 and R.Switch(root switch)



Also disconnect Switch2 and Switch3

